

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

11

Q1

ALGEBRA

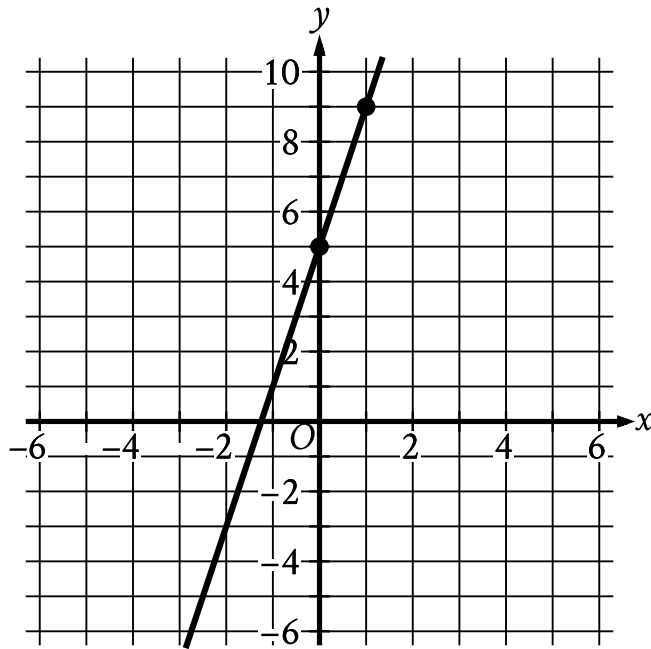
An elementary school teacher is ordering x workbooks and y sets of flash cards for a math class. The teacher must order at least 20 items, but the total cost of the order must not be over \$80. If the workbooks cost \$3 each and the flash cards cost \$4 per set, which of the following systems of inequalities models this situation?

A.
$$\begin{aligned} x + y &\geq 20 \\ 3x + 4y &\leq 80 \end{aligned}$$

B.
$$\begin{aligned} x + y &\geq 20 \\ 3x + 4y &\geq 80 \end{aligned}$$

C.
$$\begin{aligned} 3x + 4y &\leq 20 \\ x + y &\geq 80 \end{aligned}$$

D.
$$\begin{aligned} x + y &\leq 20 \\ 3x + 4y &\geq 80 \end{aligned}$$



- The line slants sharply up from left to right.
- The line passes through the following points:
 - (0 comma 5)
 - (1 comma 9)

Line j is shown in the xy -plane. Line k (not shown) is parallel to line j . What is the slope of line k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
(0)	(0)	(0)	(0)
(1)	(1)	(1)	(1)
(2)	(2)	(2)	(2)
(3)	(3)	(3)	(3)
(4)	(4)	(4)	(4)
(5)	(5)	(5)	(5)

6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

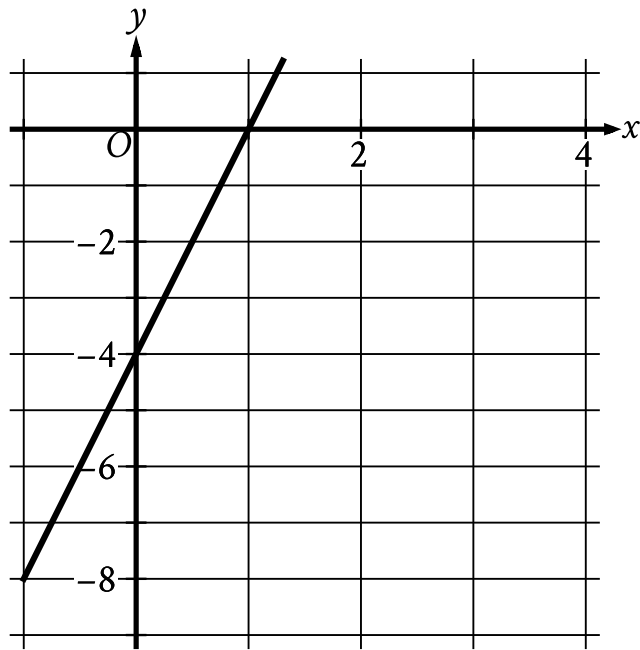
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ALGEBRA

If $4x + 2 = 12$, what is the value of $16x + 8$?

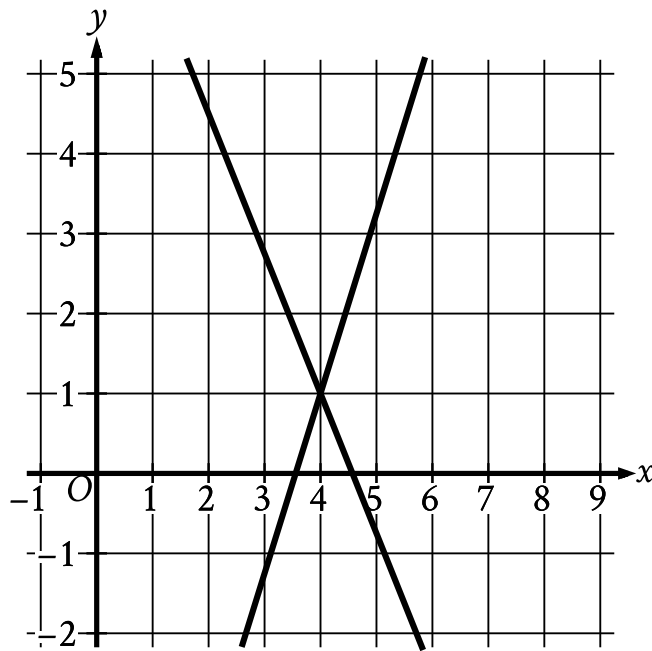
- A. 40
- B. 48
- C. 56
- D. 60



- The line slants sharply up from left to right.
- The line passes through the following points:
 - (0 comma negative 4)
 - (1 comma 0)

The graph of the function f is shown, where $y = f(x)$. What is the y-intercept of the graph?

- A. 0, -1
- B. 0, -4
- C. 0, 1
- D. 0, 4



- For the first line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (4 comma 1)
 - (StartFraction 32 Over 7 EndFraction comma 0)
- For the second line in the system:
 - The line slants sharply up from left to right.
 - The line passes through the following points:
 - (StartFraction 32 Over 9 EndFraction comma 0)
 - (4 comma 1)

The graph of a system of linear equations is shown. The solution to the system is x, y . What is the value of x ?

STUDENT-PRODUCED RESPONSE

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

ALGEBRA

Monarch butterflies can fly only with a body temperature of at least 55.0 degrees Fahrenheit °F. If a monarch butterfly's body temperature is 51.3°F, what is the minimum increase needed in its body temperature, in °F, so that it can fly?

- A. 1.3
- B. 3.7
- C. 5.0
- D. 6.3

Q7

ALGEBRA

A machine makes large boxes or small boxes, one at a time, for a total of 700 minutes each day. It takes the machine 10 minutes to make a large box or 5 minutes to make a small box. Which equation represents the possible number of large boxes, x , and small boxes, y , the machine can make each day?

- A. $5x + 10y = 700$
- B. $10x + 5y = 700$
- C. $x + y10 + 5 = 700$
- D. $10 + x5 + y = 700$

Q8

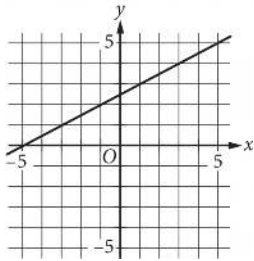
ALGEBRA

If $4x = 3$, what is the value of $24x$?

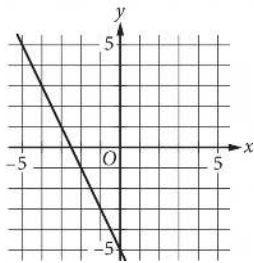
- A. $\frac{9}{2}$
- B. 6
- C. 18
- D. 72

Which of the following is the graph of the equation $y = 2x - 5$ in the xy -plane?

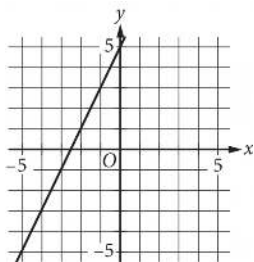
A.



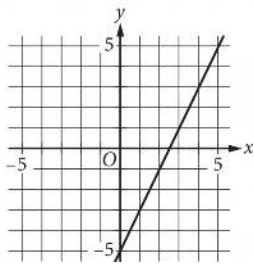
B.



C.



D.



Q10

ALGEBRA

$$5x = 15$$

$$-4x + y = -2$$

The solution to the given system of equations is x, y . What is the value of $x + y$?

- A. -17
- B. -13
- C. 13
- D. 17